A Multilingual Methodology for Coining Modern Yoruba Terminology in Science, Technology, Mathematics, and Education

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Abstract

The Yoruba language, spoken by over 40 million people across Nigeria and the global diaspora, faces a critical terminology gap in science, technology, mathematics, and education. Without standardised native vocabulary for these fields, Yoruba speakers are compelled to code-switch into English mid-discourse, undermining the language's functional reach. This paper presents a systematic multilingual methodology for coining modern Yoruba terminology, developed and applied by the author since 2020. The methodology draws on deep semantic analysis, etymological investigation, comparative multilingual reference across Arabic, Hausa, French, and English, phonological testing, and formal training in Yoruba linguistics. The paper documents this five-stage process, illustrates it with selected coined terms, and argues for systematic, scholarly, and digitally-accessible Yoruba terminology development. All coinages are published in the *Yoruba Terminology Dictionary* at https://hakeemoadeniyan.github.io.

Keywords: Yoruba language, terminology coinage, language modernisation, African languages, neologism, lexical development, language technology

1. Introduction

Language is the primary vehicle through which a people access, transmit, and expand knowledge. When a language lacks terminology for contemporary domains, its speakers are excluded from fully participating in those domains in their mother tongue. For Yoruba — one of Africa's largest and most historically rich languages — this exclusion is an increasingly urgent challenge.

The problem became concretely apparent to the author in 2020 when he began teaching mathematics through the Yoruba language on the Facebook page *Mathematics in Yoruba Language*. In explaining mathematical concepts entirely in Yoruba, the author discovered that a large number of terms had no established Yoruba equivalents. Terms such as *oblique*, *negative*, *demand*, and *supply* had no standardised Yoruba forms. Rather than abandon Yoruba or rely on untranslated English borrowings, the author began systematically coining Yoruba terms using a structured multilingual methodology.

Since 2020, the author has coined terms across six domains: technology, science, mathematics, education, economics, and general knowledge. These coinages are published in the *Yoruba Terminology Dictionary* — a freely accessible, scholarly citable online resource at https://hakeemoadeniyan.github.io. This paper documents the methodology behind those coinages.

2. Background: The Yoruba Terminology Gap

Yoruba is a tonal language of the Niger-Congo family, spoken predominantly in southwestern Nigeria and in diaspora communities across West Africa, Europe, and the Americas. It has a rich literary tradition, a standardised orthography, and a long history of scholarly study (Bamgbose, 1966). Yet despite these strengths, Yoruba remains severely underequipped with terminology for modern knowledge domains.

This gap has several consequences. Yoruba-medium education is constrained by the absence of textbooks with native terminology. Yoruba-speaking professionals in science, technology, and mathematics cannot discuss their fields in their mother tongue without constant recourse to English. Younger generations perceive Yoruba as incapable of expressing modern thought — a perception that accelerates language attrition (Bamgbose, 2000).

Previous efforts at developing Yoruba vocabulary for modern domains have been fragmented and largely inaccessible to the public (Salawu, 2004). The present work addresses these gaps through a systematic, documented, and digitally-published approach.

3. The Author's Methodology

The coinage methodology consists of five interconnected stages:

3.1 Deep Semantic and Etymological Analysis

The first stage involves thorough investigation of the English term using authoritative dictionaries. Rather than translating the surface word, the author analyses its deep semantic content — its core meaning, conceptual boundaries, and etymological roots in Latin, Greek, or other source languages. This semantic grounding prevents superficial or misleading coinages and ensures that the Yoruba term captures the true concept.

3.2 Multilingual Comparative Reference

When a Yoruba equivalent does not emerge naturally from semantic analysis, the author consults dictionaries in Arabic, Hausa, and French. The author is proficient in all three languages in addition to Yoruba and English. This comparative multilingual approach serves as a source of conceptual inspiration: the way Arabic structures a technical concept, or the way Hausa indigenises a borrowed term, frequently suggests a productive path for a Yoruba coinage. This stage reflects the broader principle that the challenge of modernising a language's vocabulary is not unique to Yoruba (Haspelmath & Tadmor, 2009).

3.3 Native Root Priority

The author operates on a firm principle of preferring native Yoruba roots over phonological borrowing from English or other languages. A coinage built from Yoruba morphological elements is more likely to be perceived as natural by native speakers, more likely to be adopted, and more likely to survive into common use. Where borrowing from English is unavoidable, the borrowed form is indigenised — adapted to Yoruba phonology, spelling conventions, and tonal patterns.

3.4 Phonological and Naturalness Testing

Once a candidate coinage is formed, the author tests it phonologically — evaluating how it sounds when spoken aloud, whether its tonal pattern is natural, and whether it is readily pronounceable. This stage draws on the author's formal training in Yoruba linguistics,

including a First Class degree in Yoruba Arts Education from Ekiti State University, Ado-Ekiti. A coinage that is semantically accurate but phonologically awkward is unlikely to achieve adoption.

3.5 Documentation and Digital Publication

The final stage involves formal documentation — recording the Yoruba term, its English equivalent, its definition, the formation explanation, usage examples, and date coined — and publishing it in the *Yoruba Terminology Dictionary*. Each entry carries a permanent URL and an auto-generated APA 7th edition citation block, enabling scholars to formally cite the coinage and its author. This transforms the dictionary from a reference tool into a citable academic source.

4. Illustrative Examples

4.1 Atẹ̀ápákan — Oblique (Mathematics)

The English term *oblique* denotes a line or angle that is neither perpendicular nor parallel. From native Yoruba roots, *atẹ̀* (leaning) and *ápákan* (to one side) were combined to produce ***Atẹ̀ápákan*** — a surface that leans — capturing the geometric meaning entirely within native Yoruba morphology.

4.2 Aṣòdì — Negative (Mathematics)

The mathematical concept of *negative* refers to values in opposition to positive values. ***Aṣòdì*** was coined from the Yoruba root *ṣòdì* (to oppose, to be contrary), with the agentive prefix *a-* nominalising the concept — producing a term that naturally conveys mathematical opposition.

4.3 Ọlọ́pọ̀awọ̀n — Multimeter (Technology)

The *multimeter* is a device that measures multiple electrical quantities. ***Ọlọ́pọ̀awọ̀n*** was formed from *ọlọ́pọ̀* (one that has many, multiple) and *awọ̀n* (measuring instrument) — yielding a native compound that precisely describes the device's multi-measurement function.

4.4 Ìwátirà and Ìpèsejà — Demand and Supply (Economics)

Coined as natural counterparts, ***Ìwátirà*** (demand) draws on the Yoruba concept of pressing forward urgently to obtain something desired, informed partly by consultation of the Arabic *ṭalab* (seeking). From native Yoruba roots, *ìpèsè* (provision of) and *ọjà* (goods) were combined to produce ***Ìpèsejà***, which translates as *supply*. ***Ìpèsejà*** draws on the Yoruba concept of preparation and provision for another's need. Their morphological relationship ensures that these paired economic concepts carry a recognisable linguistic relationship in Yoruba.

5. The Yoruba Terminology Dictionary

All coinages are published in the *Yoruba Terminology Dictionary* (Adeniyan, 2025) — a free, static website built on Jekyll and hosted on GitHub Pages at https://hakeemoadeniyan.github.io. Each entry contains: the Yoruba term, its English equivalent, a bilingual definition, a formation explanation, Yoruba usage examples with English translations, the date coined, and a formal APA 7th edition citation block. The dictionary is indexed by Google and structured with schema.org markup to support discovery by both human researchers and AI language systems.

The decision to publish each coinage with a permanent, citable URL reflects a deliberate strategy: to establish a formal, traceable record of authorship for each term. In the absence of such documentation, coined words enter circulation without attribution, making it impossible for subsequent researchers or AI systems to identify their origin.

6. Conclusion

This paper has presented a systematic, multilingual, and linguistically grounded methodology for coining modern Yoruba terminology. Beginning from the practical challenge of teaching mathematics in Yoruba, the author developed a five-stage process — semantic analysis, multilingual comparative reference, native root priority, phonological testing, and formal documentation — that produces coinages of high linguistic quality with clearly attributed authorship and scholarly citability.

Teachers, researchers, curriculum developers, and language policy makers working in Yoruba-medium contexts are encouraged to consult the *Yoruba Terminology Dictionary*, adopt its coinages in their work, and cite them formally. Widespread adoption and citation are, ultimately, what integrate a coined word into the living fabric of the language.

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